RainGuard HK Rainfall Prediction and Weather Decision Support

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1 Research Problem and Objectives

Hong Kong facility managers must combine rainfall observations, forecasts, coastal tide readings and operational guidance from separate official sources. A weather dashboard alone does not establish which evidence supports a decision. We propose RainGuard HK, a decision-support prototype that connects rainfall classification with source-cited weather explanations and human-reviewed operational records. Our research asks: *Can simple machine-learning models identify next-day high-rainfall events, and does adding current observations to retrieved official guidance improve the factual quality of operational answers?*

The objectives are to (1) compare interpretable rainfall classifiers using chronological evaluation, (2) assess an assistant grounded in official guidance, with provenance and freshness checks, and (3) demonstrate a traceable workflow for a real Hong Kong facility, from observation to review and shift report. The system will supplement official services; it will not issue weather warnings or claim to predict site flooding.

2 Proposed Data and Methodology

Data. We have obtained Hong Kong Observatory (HKO) daily rainfall records for HKO headquarters, Lau Fau Shan and Sha Tin, covering January 2000–August 2026: 9,740 dated entries per station before quality filtering [1]. Live inputs comprise 36 station rainfall readings, 18 district rainfall summaries, seven tide stations, official warnings, two-hour gridded rainfall nowcasts and local forecasts. Seven HKO, Drainage Services Department and Labour Department webpages provide the external knowledge corpus [2–4]. Raw files, timestamps and SHA-256 hashes will document provenance. Missing and incomplete readings will retain quality flags; trace-rain handling and the number of excluded samples will be reported.

Prediction task. At the end of day t, predict whether station rainfall on day $t+1$ is at least 25 mm, a project-defined label rather than an official warning threshold. Predictors use only information available by t: rainfall lags, seven-day summaries and seasonal features. We will compare logistic regression and random forest with constant-prevalence and persistence baselines. Training uses 2000–2019, validation 2020–2022 and a locked test set 2023–2025; 2026 records are excluded from model selection. Preprocessing and thresholds will be fitted without test data. We will report PR-AUC, ROC-AUC, recall, F1 and Brier score by station, with a trace-rain sensitivity analysis. Classification and regression concepts connect directly to Lectures 4 and 5.

Knowledge assistant and evaluation. Motivated by retrieval-augmented generation [5], we extract and chunk official text, then rank passages by keyword and Chinese character-pair matching. DeepSeek combines retrieved passages and observations from the server to produce cited answers, without fine tuning. We will compare three answer conditions on 30 predefined questions: no grounding, official text only, and official text with observations. Two reviewers will independently assess factual support, citation correctness and recognition of stale or missing data, then reconcile disagreements. Official forecasts, model outputs and manual rules will be displayed separately.

3 Expected Output and Team Responsibilities

We will deliver reproducible experiments and a web application in simplified Chinese, including regional comparisons, rainfall window screening, facility review records and handover reports. Screenshots and a walkthrough will use actual observations and explicitly identified historical snapshots. Facility records require manual verification; browser-local storage is a prototype limitation. No flood-probability or safety guarantee will be inferred from tide height or district rainfall.

Table 1: Proposed responsibilities for six team members.

Member information	Responsibility and output
M1 Name: _____ ID: _____	Data collection, provenance and quality audit.
M2 Name: _____ ID: _____	Features and trace-rain sensitivity analysis.
M3 Name: _____ ID: _____	Model training and baseline evaluation.
M4 Name: _____ ID: _____	Knowledge corpus, retrieval and assistant.
M5 Name: _____ ID: _____	Interface, facility workflow and report export.
M6 Name: _____ ID: _____	Answer evaluation, testing and presentation.

References

[1] Hong Kong Observatory. Open Data: climate records and weather APIs. Accessed October 2026.
[2] Hong Kong Observatory. Rainstorm Warning System. Accessed October 2026.
[3] Drainage Services Department. How to Reduce Flood Damage. Accessed October 2026.
[4] Labour Department. Work Arrangements in Adverse Weather. Accessed October 2026.
[5] Patrick Lewis et al. 2020. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. NeurIPS.